Derivation of the BFKL Term in NLO Gluon Production

How the $\log(\vee/k^+)$ term with the $K_{ww'}(z)\,K_{xx'}(z)$ structure follows from the $(1-\xi)/\xi$ term of the splitting function

$d_\perp = 2-\epsilon$, $\overline{\text{MS}}$ scale $\mu \equiv \mu_{\overline{MS}}$, $c_0 \equiv 2e^{-\gamma_E}$, transverse measure $\int_x \equiv \int d^2x$.

1. Goal

Start from the first line of the NLO spectrum,

$$\frac{d^3N^{(\mathrm{I})}}{d^3k} = \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1\Big]$$
$$\times\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)/\xi}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big]\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ec}(x)\big]\rho^{e'}(x')\rho^e(x),$$

and show that its $\log(\vee/k^+)$-enhanced part is exactly

$$\frac{d^3N^{\mathrm{BFKL}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\times\big[U^{e'c}(z)-U^{e'c}(x')\big]\big[U^{ec}(z)-U^{ec}(x)\big]\rho^{e'}(x')\rho^e(x).$$

> The last Wilson line is written $U^{ce}(x)$ in both of the original expressions. It is taken here to be $U^{ec}(x)$, the same in both, so it does not affect the derivation.

2. Notation

$$C \equiv \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+},\qquad a\equiv\frac{k^+}{\vee},\qquad r\equiv y'-y,$$

$$\mathcal{A}(x,y,x',y')\equiv\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2},\qquad \mathcal{O}(y,y')\equiv\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ec}(x)\big]\rho^{e'}(x')\rho^e(x),$$

$$K_{ww'}(z)\equiv\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}.$$

The dependence of $\mathcal{O}$ on x,x' is kept implicit. In this notation the target result reads

$$\frac{d^3N^{\mathrm{BFKL}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,K_{ww'}(z)\,\mathcal{A}(x,z,x',z)\,\mathcal{O}(z,z).$$

3. Step 1: expand the bracket

Call the square bracket in the first line $L(r)$. Pull out $d_\perp/2$ and use γ_E to define c_0:

$$L=-\frac{d_\perp}{2}\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2}{4}\Big]-1=-\frac{d_\perp}{2}\Big[\frac{2}{\epsilon}+\log\frac{r^2\mu^2}{c_0^2}\Big]-1,\qquad \log\frac{r^2\mu^2e^{2\gamma_E}}{4}=\log\frac{r^2\mu^2}{c_0^2}.$$

Insert $d_\perp = 2-\epsilon$:

$$L=-\Big(1-\frac{\epsilon}{2}\Big)\Big[\frac{2}{\epsilon}+\log\frac{r^2\mu^2}{c_0^2}\Big]-1=-\frac{2}{\epsilon}+1-\log\frac{r^2\mu^2}{c_0^2}-1+O(\epsilon)=-\frac{2}{\epsilon}-\log\frac{r^2\mu^2}{c_0^2}+O(\epsilon).$$

For what follows, split L into a part that does not depend on r and a part that does:

$$L(r)=\underbrace{\Big[-\frac{2}{\epsilon}-\log\frac{\mu^2}{c_0^2}\Big]}_{\equiv L_0\ (\text{no } r)}\ \underbrace{-\log r^2}_{r\text{-dependent}}.$$

Each of the two logarithms on the right carries a compensating unit; only their sum is meaningful, and nothing below depends on that unit.

4. Step 2: which term of the splitting function can give $\log(\vee/k^+)$

A $\log(\vee/k^+)=\log(1/a)$ can only come from the lower limit $\xi\to a\to 0$. Define

$$\Phi_W(q)\equiv\int_{x,y,x',y'}e^{-iq\cdot(y'-y)}\,W\,\mathcal{O}(y,y').$$

The first line is then

$$\frac{d^3N^{(\mathrm{I})}}{d^3k} = C\int_a^1 \frac{d\xi}{\xi^2}\Big[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\Big]\,\Phi_{\mathcal{A}L}\Big(\frac{k}{\xi}\Big).$$

As $\xi \to 0$, the momentum $k/\xi \to \infty$, so what decides whether a log appears is the large-q behaviour of $\Phi_{\mathcal{A}L}(q)$. Section 5 shows

$$\Phi_{\mathcal{A}L}(q) \xrightarrow{q\to\infty} \frac{\text{const}}{q^2} \qquad \Longrightarrow \qquad \Phi_{\mathcal{A}L}(k/\xi) \sim \xi^2.$$

The three terms of the splitting function therefore behave as follows:

Term	**Weight $\frac{1}{\xi^2}\times$ term**	**$\times\,\xi^2$ from Φ**	**$\int_a d\xi$ as $a\to 0$**
$\xi(1-\xi)$	$\frac{1-\xi}{\xi}$	$\xi(1-\xi)$	finite, no log
$\frac{1-\xi}{\xi}$	$\frac{1-\xi}{\xi^3}$	$\frac{1-\xi}{\xi}$	$\log(1/a) = \log(\vee/k^+)$
$\frac{\xi}{(1-\xi)_+}$	$\frac{1}{\xi(1-\xi)_+}$	$\frac{\xi}{(1-\xi)_+}$	finite, no log

The subtraction in the plus distribution acts only at $\xi = 1$, so it cannot produce a log at $\xi \to 0$. Only the $(1-\xi)/\xi$ term is needed:

$$T \equiv C\int_a^1 d\xi\,\frac{1-\xi}{\xi^3}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)/\xi}\,\mathcal{A}(x,y,x',y')\,L(y'-y)\,\mathcal{O}(y,y').$$

5. Step 3: the r-independent part L_0 gives no log

The L_0 part of T is

$$T_0 = C\,L_0\int_a^1 d\xi\,\frac{1-\xi}{\xi^3}\,\Phi_{\mathcal{A}}\Big(\frac{k}{\xi}\Big).$$

$\Phi_{\mathcal{A}}(q)$ is the LO structure at transverse momentum q (amplitude times conjugate):

$$\Phi_{\mathcal{A}}(q) = a_i(q)\,a_i^\dagger(q), \qquad a_i(q) = \int_{x,y} e^{iq\cdot y}\frac{(x-y)_i}{(x-y)^2}\big[U(y)-U(x)\big]\rho(x).$$

At large q, a_i is controlled by $y\to x$. There $[U(y)-U(x)] \simeq (y-x)_j\partial_j U(x)$ vanishes linearly and cancels the $1/|x-y|$ of the kernel. What is left, $(x-y)_i(y-x)_j/(x-y)^2$, is bounded, so its Fourier transform falls like $1/q^2$. Hence

$$a_i \sim \frac{1}{q^2}, \qquad \Phi_{\mathcal{A}}(q) \sim \frac{1}{q^4},$$

which is the familiar $1/k^4$ tail of the LO spectrum. Therefore

$$\frac{1-\xi}{\xi^3}\,\Phi_{\mathcal{A}}(k/\xi) \sim \xi \qquad \Longrightarrow \qquad T_0 \text{ has a finite limit as } a\to 0.$$

So the collinear pole $-2/\epsilon$ and the μ-dependence do **not** enter the BFKL term, as the target result requires. Only the $-\log r^2$ part remains:

$$T\big|_{\log} = C\int_a^1 d\xi\,\frac{1-\xi}{\xi^3}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)/\xi}\,\big[-\log(y'-y)^2\big]\,\mathcal{A}(x,y,x',y')\,\mathcal{O}(y,y').$$

6. Step 4: rescale the dipole

At fixed x, x', y, change variables from y' to s:

$$y' = y+\xi s, \qquad d^2y' = \xi^2\,d^2s, \qquad \frac{k\cdot(y'-y)}{\xi} = k\cdot s, \qquad (y'-y)^2 = \xi^2 s^2.$$

Then

$$T\big|_{\log} = C\int_a^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',s} e^{-ik\cdot s}\,\big[-\log s^2 - \log\xi^2\big]\,\mathcal{A}(x,y,x',y+\xi s)\,\mathcal{O}(y,y+\xi s).$$

The ξ^2 from the Jacobian has turned $1/\xi^3$ into $1/\xi$.

The $-\log\xi^2$ piece gives no log. It has no s-dependence, so undoing the rescaling gives $-\log\xi^2\cdot\xi^{-2}\Phi_{\mathcal{A}}(k/\xi) \sim \xi^2\log\xi$. Its integrand behaves as $\xi\log\xi$, which is integrable at $\xi = 0$.

What remains is

$$T\big|_{\log} = C\int_a^1 d\xi\,\frac{1-\xi}{\xi}\,F(\xi) + (\text{finite as } a\to 0), \qquad F(\xi) \equiv \int_{x,y,x',s} e^{-ik\cdot s}\big(-\log s^2\big)\,\mathcal{A}(x,y,x',y+\xi s)\,\mathcal{O}(y,y+\xi s).$$

7. Step 5: take $\xi \to 0$ inside F

Write $F(\xi) = F(0) + [F(\xi) - F(0)]$. For fixed x, x', y, the product $\mathcal{A}\mathcal{O}$ is smooth in y' near $y' = y$, so $F(\xi) - F(0) = O(\xi)$. Then $\int_a^1 d\xi\,\frac{1-\xi}{\xi}[F(\xi) - F(0)]$ has a finite limit as $a \to 0$ and gives no log. At $\xi = 0$, $y' = y$; rename $y \equiv z$:

$$F(0) = \int_{x,x',z} \mathcal{A}(x,z,x',z)\,\mathcal{O}(z,z)\;\times\; J(k), \qquad J(k) \equiv \int d^2s\, e^{-ik\cdot s}\big(-\log s^2\big),$$

$$\mathcal{A}(x,z,x',z) = \frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}, \qquad \mathcal{O}(z,z) = \big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x).$$

> **Physical picture.** $\xi = k^+/p^+ \to 0$ means the parent gluon is much harder than the measured one. The rescaling shows that its transverse size $|y' - y| = \xi|s|$ shrinks to zero. The hard gluon therefore crosses the target at one point z in both the amplitude and the conjugate, while the soft measured gluon carries the momentum k. This is exactly the structure of the target result.

8. Step 6: Fourier transform of the logarithm

In two dimensions, $\log|s|$ is the Green's function of the Laplacian:

$$\nabla_s^2 \log s^2 = 2\,\nabla_s^2 \log|s| = 4\pi\,\delta^2(s).$$

Multiply by $e^{-ik\cdot s}$ and integrate over s. On the left, integrate by parts twice ($\nabla^2 e^{-ik\cdot s} = -k^2 e^{-ik\cdot s}$):

$$-k^2\int d^2s\, e^{-ik\cdot s}\log s^2 = 4\pi \qquad\Longrightarrow\qquad \int d^2s\, e^{-ik\cdot s}\log s^2 = -\frac{4\pi}{k^2} \quad (k\neq 0).$$

This holds in the distributional sense; any additive constant in $\log s^2$ contributes only $\delta^2(k)$, which vanishes for $k \neq 0$. Therefore

$$J(k) = \int d^2s\, e^{-ik\cdot s}\big(-\log s^2\big) = \frac{4\pi}{k^2}.$$

The same holds with $e^{+ik\cdot s}$, since $\log s^2$ is even in s.

9. Step 7: write $4\pi/k^2$ as the soft-gluon kernel

(a) One leg. Use $\frac{u_i}{u^2} = \frac{1}{2}\partial_i \log u^2$ and integrate by parts:

$$I_i(k) \equiv \int d^2u\, e^{ik\cdot u}\frac{u_i}{u^2} = \frac{1}{2}\int d^2u\, e^{ik\cdot u}\,\partial_i \log u^2 = -\frac{ik_i}{2}\int d^2u\, e^{ik\cdot u}\log u^2 = -\frac{ik_i}{2}\Big(-\frac{4\pi}{k^2}\Big) = 2\pi i\,\frac{k_i}{k^2}.$$

Since u_i/u^2 is odd, $I_i(-k) = -I_i(k) = -2\pi i\,k_i/k^2$.

(b) Both legs. For any fixed z, put $u = w - z$, $u' = w' - z$. Then $w' - w = u' - u$, and the integral factorizes:

$$\int_{w,w'} e^{-ik\cdot(w'-w)}K_{ww'}(z) = \int_u e^{ik\cdot u}\frac{u_i}{u^2}\int_{u'} e^{-ik\cdot u'}\frac{u'_i}{u'^2} = I_i(k)\,I_i(-k) = \Big(2\pi i\frac{k_i}{k^2}\Big)\Big(-2\pi i\frac{k_i}{k^2}\Big) = \frac{4\pi^2}{k^2}.$$

Comparing with Step 6:

$$J(k) = \frac{4\pi}{k^2} = \frac{1}{\pi}\int_{w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2} \qquad \text{for any } z.$$

The point z can be taken to be the position of the hard gluon, which is what turns $J(k)$ into the emission kernel of the soft measured gluon at w, w'.

10. Step 8: the ξ integral

$$\int_a^1 d\xi\,\frac{1-\xi}{\xi} = \Big[\log\xi - \xi\Big]_a^1 = \log\frac{1}{a} - 1 + a = \log\frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}.$$

Keep $\log(\vee/k^+)$. The $-1 + k^+/\vee$ is a non-logarithmic constant and belongs in the finite remainder.

11. Step 9: assemble

Collecting Steps 2–8,

$$T\big|_{\log(\vee/k^+)} = C\,\log\frac{\vee}{k^+}\,F(0) = C\,\log\frac{\vee}{k^+}\int_{x,x',z}\mathcal{A}(x,z,x',z)\,\mathcal{O}(z,z)\,\frac{1}{\pi}\int_{w,w'}e^{-ik\cdot(w'-w)}K_{ww'}(z).$$

The prefactor is

$$\frac{C}{\pi} = \frac{g^2}{4\pi^2}\cdot\frac{N_c}{(2\pi)^3}\cdot\frac{g^2}{2\pi^2}\cdot\frac{1}{k^+}\cdot\frac{1}{\pi} = \frac{g^4N_c}{(4\pi^2)(2\pi^2)\,\pi}\,\frac{1}{(2\pi)^3}\frac{1}{k^+} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}.$$

Therefore

$$\frac{d^3N^{\mathrm{BFKL}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\times\,\big[U^{e'c}(z)-U^{e'c}(x')\big]\big[U^{ec}(z)-U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x),$$

which is the target result, including the prefactor. Everything dropped along the way has a finite limit as $\vee\to\infty$.

12. Summary of what was dropped, and why

Piece	Small-ξ behaviour of the integrand	Why no $\log(\vee/k^+)$
$\xi(1-\xi)$ and $\xi/(1-\xi)_+$ terms	$\sim\xi$	Section 4
Pole $-2/\epsilon$ and $\log\mu^2$ (L_0)	$\sim\xi$	LO tail $\Phi_{\mathcal{A}}(q)\sim q^{-4}$ (Section 5)
$-\log\xi^2$ from the rescaling	$\sim\xi\log\xi$	integrable (Section 6)
$F(\xi)-F(0)$	$O(1)$	$\int_0 d\xi\,O(\xi)/\xi$ converges (Section 7)
$-1+k^+/\vee$ from $\int d\xi(1-\xi)/\xi$	—	constant (Section 10)

13. The other two lines of the NLO spectrum

13.1 The third line (the $d_\perp$-weighted ξ integral) gives no $\log(\vee/k^+)$

Its weight is $d\xi\,\frac{1-\xi}{\xi}$, with no $1/\xi^2$, and its integrand is $\mathcal{A}-2\mathcal{B}$, where $\mathcal{B}=\frac{(x-y)\cdot r\,(x'-y')\cdot r}{(x-y)^2(x'-y')^2r^2}$. As $r\to0$, $(\mathcal{A}-2\mathcal{B})\propto(\delta_{ij}-2\hat r_i\hat r_j)$, which is traceless. Its Fourier transform is $-\frac{4\pi}{q^2}(\delta_{ij}-2\hat q_i\hat q_j)$, so $\Phi_{\mathcal{A}-2\mathcal{B}}(k/\xi)\sim\xi^2$ and the integrand goes like ξ. There is no log.

13.2 The second line: the explicit $\log(\vee/k^+-1)$

With $b\equiv k^+/(k^++\Lambda)$, $b/(1-b)=k^+/\Lambda$ and $a/(1-a)=k^+/(\vee-k^+)$ hold exactly, the blue bracket of the second line is exactly

$$\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)=\int_{k^+/\vee}^{k^+/(k^++\Lambda)}d\xi\Big[\frac{1}{1-\xi}+\frac{1}{\xi}\Big].$$

Its $\int d\xi/\xi$ part multiplies $\Phi_{\mathcal{A}L}(k)$: the phase is frozen at its $\xi=1$ value, the Wilson lines sit at $y\neq y'$, and the collinear pole $-2/\epsilon$ is still present. That is not the $\xi\to0$ limit of the $(1-\xi)/\xi$ term, which is the structure derived above.

At large k it tends to the same value as the BFKL term, so keeping both would double it. To end up with only the target BFKL term, the piece $C\log(\vee/k^+)\,\Phi_{\mathcal{A}L}(k)$ must cancel against contributions not shown, for example virtual graphs with the same cutoff $\vee$. Otherwise it is double counting from the step that produced $\log(\vee/k^+-1)$.